

Grounded Exploration: Decoupled Planning and Terrain-Adaptive Execution from Onboard World Feedback

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Abstract

Classical deterministic planners fail in unstructured real-world environments where maps and external instrumentation are unavailable. We present a real-world reinforcement learning pipeline for a physical rover that learns to explore using world feedback alone. Intrinsic reward is derived from a Vision Memory Model that scores how novel each new scene is using Random Network Distillation on live camera frames. We deploy and train directly on hardware with a safety layer constraining motor commands, and propose novel places found — the number of visually distinct scenes accumulated during an evaluation episode — as a practical, instrumentation-free exploration metric.

Placeholder Figure 1: Mapless real-rover RLxF exploration stack

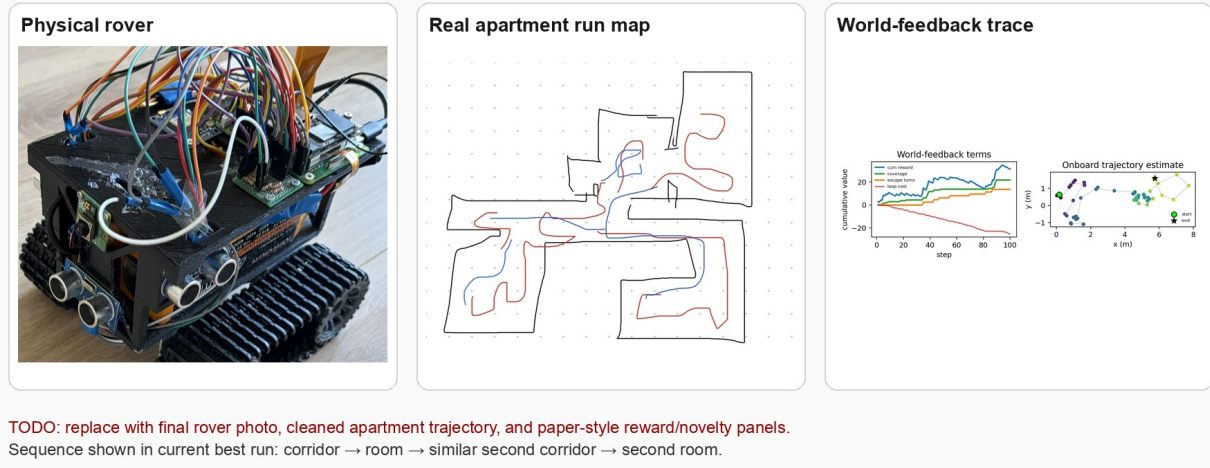


Figure 1: Placeholder — get better images of this. Real-rover RLxF exploration overview: physical platform, apartment-scale trajectory, and world-feedback reward traces. This temporary teaser sets the intended first-page feel before final photography and plotting.

1. Introduction

Autonomous exploration is usually framed as mapping: estimate free space, detect frontiers, and plan to informative boundaries (Yamauchi, 1997; Thrun et al., 2005). This assumption is brittle on small low-cost robots with only monocular vision, short-range sensors, and noisy inertial measurements. In damaged

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buildings or rubble, the map may also change while the robot is operating.

We study mapless exploration from world feedback. A physical rover learns online without external instrumentation, pre-built maps, or simulator pretraining. After each action, the only feedback is onboard: camera frames, range sensors, inertial response, executed motion, and recovery events. The environment supplies consequences; the designer specifies how to turn them into reward.

The central hypothesis is that learning should decide where to try next, not raw motor PWM. A deterministic executor realizes the command near obstacles. Our

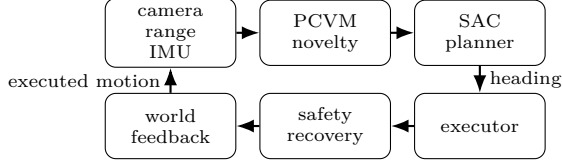


Figure 2. Current decoupled architecture. The Predictive Contextual Visual Memory (PCVM) module supplies novelty; SAC chooses a relative heading; the executor and safety layer determine what actually happens and return world feedback.

strongest current abstraction is a relative heading followed by forward motion until the world, via front-range feedback, says to stop.

Intrinsic motivation is a natural reward source (Pathak et al., 2017; Burda et al., 2019; Savinov et al., 2019), but real hardware exposes failure modes hidden in simulation: blur can look novel, wall contact can produce new pixels, and corridor loops can collect reward. We therefore gate novelty by executed motion and log penalties for recovery, zero-progress, and revisits.

This is an in-progress real-rover RLxF report. The policy is trained on hardware; saved trajectories are for auditing and ablations, not supervised pretraining. Evaluation uses quantities available to the robot and reviewer: successful motion, novel places, recovery frequency, and qualitative apartment coverage.

Our contribution is a minimal real-world benchmark and prototype stack: onboard RL, intrinsic visual memory, safety-gated local actions, and auditable world-feedback rewards.

2. Methodology

2.1. Problem setting

At time t the rover receives

$$o_t = (I_t, d_t, m_t, a_{t-1}, e_{t-1}), \quad (1)$$

where I_t is the camera image, d_t range, m_t inertial/proprioceptive state, a_{t-1} the previous high-level action, and e_{t-1} what the executor actually achieved. The policy selects a local guide a_t , not motor commands; the executor applies safety constraints and returns world feedback.

The goal is exploration under a fixed step budget, without a metric map or ground-truth pose. Dead-reckoned pose is only diagnostic.

2.2. Local action abstraction

The current action is a scalar relative heading

$$a_t \in [-1, 1], \quad \theta_t = \theta_{\max} a_t, \quad (2)$$

with $\theta_{\max} = 75^\circ$. The executor turns by θ_t , then drives until the front sensor reaches a threshold or a max duration expires. This local, egocentric interface avoids global localization and discourages tiny dithering moves.

Other guides, such as short (θ, ρ) targets, fit the same framework. The key separation is intent from execution.

2.3. Onboard novelty memory

We call the current memory backend Predictive Contextual Visual Memory (PCVM). It maps the frame and action context to z_t and stores representative embeddings. Novelty is nearest-memory distance and/or RND-style prediction error (Burda et al., 2019):

$$n_t = \alpha \text{dist}(z_t, \mathcal{M}_{t-1}) + \beta \|f_{\text{pred}}(I_t) - f_{\text{target}}(I_t)\|_2^2, \quad (3)$$

where $\mathcal{M}_{t-1}$ is memory and f_{target} is fixed. Distant embeddings create or update memory representatives.

The final encoder is not fixed; we have tested lightweight CNNs, recurrent path-conditioned memory, and frozen backbones. The framework only requires onboard novelty estimates.

2.4. World-feedback reward

Novelty alone is insufficient: the robot can see new pixels while stuck. Reward combines perceptual novelty with execution feedback:

$$r_t = r_t^+ - r_t^-, \quad (4)$$

$$r_t^+ = w_n \mathcal{K}[M_t] n_t + w_c C_t + w_d \Delta s_t + w_s S_t, \quad (5)$$

$$r_t^- = \lambda_r R_t + \lambda_0 Z_t + \lambda_l L_t + \lambda_o O_t. \quad (6)$$

Here M_t gates novelty by motion, C_t denotes cluster/coverage expansion, S_t safe motion, Δs_t executed distance, and R_t, Z_t, L_t, O_t recovery, zero-forward, revisit, and obstacle penalties.

Each term is logged separately, so scalar reward can be audited against video. This exposed early over-rewarding of corridor loops and motivated loop-aware penalties.

2.5. Training and data usage

We currently use SAC (Haarnoja et al., 2018). Replay is populated online from rover experience. Safety can

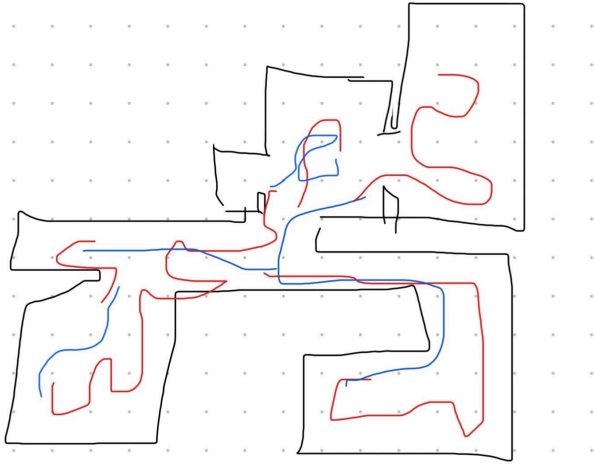


Figure 3. Placeholder apartment/run map. This half-column figure is only for layout exploration; replace with a cleaned schematic or manually annotated map before submission.

tact/stall events and many recoveries, so the controller is not solved; the result only suggests that the action/reward interface can produce room-to-room exploration.

clip execution, and the policy receives the executed-action feedback.

Saved manual/autonomous runs are used only for reward auditing, memory diagnostics, and figures, not policy pretraining.

2.6. Evaluation protocol

Evaluation uses short autonomous episodes with successful-forward fraction, recoveries, zero-progress, dead-reckoned spread, and frame review. Frozen runs serve as reward-alignment diagnostics.

3. Preliminary Real-Rover Experiments

Use case framing. The motivating deployment is an unknown and changing indoor structure, e.g. a small rescue rover entering damaged buildings or rubble. The robot cannot rely on a precomputed map, but it can evaluate the consequences of its actions through onboard feedback.

Representative run. Figure 1 includes reward and trajectory panels from a 100-step continuation run. The human-observed sequence was corridor $\rightarrow$ room $\rightarrow$ visually similar second corridor $\rightarrow$ second room. The third segment receives less novelty, plausibly because the apartment has an X-shaped corridor geometry with overlapping views. This is qualitative until repeated runs and ablations are complete.

The run scored +31.5 over 100 actions, traveled 20.3 m, reached 5.7 m net displacement, and expanded the dead-reckoned box to 22.3 m². It still had two con-

Table 1. Placeholder ablation and benchmark matrix. Empty cells mark measurements to fill after repeated runs.

System / ablation	Rooms	Travel	Contacts	Reward	Notes	Status
Short local target PCVM						old baseline
Theta-front PCVM						current baseline
Theta-front + coverage reward						current run
No surprise term						planned
Executed-action memory						planned
Human WASD reference						reference

4. Ablations

Draft notes from current rover runs. Current evidence points first to action interface: short local targets produced timid motion, while relative-heading plus world-bounded execution reached multiple rooms. This motivates a fixed-distance target vs. drive-until-front ablation. A second ablation is requested-action memory vs. executed-action memory, because safety clipping and recovery change what physically happened. A third is surprise on/off: transition error may capture useful unmodeled outcomes, but may also reward blur, noise, or collision artifacts.

Reward-alignment drafts. The coverage reward was introduced after visual review showed that a two-room run scored lower than a corridor loop. Future ablations should rank frozen runs by human labels: corridor loop < corridor+one-room < corridor+two-room. This allows reward tuning before spending hardware time.

Architecture drafts. The current PCVM module still aliases similar rooms/corridors. We plan to compare lightweight CNN features, frozen pretrained encoders, and dual path/visual memory while keeping the same theta-front executor.

5. Conclusion & Future Work

This draft supports a narrow claim: decoupled planning and terrain-adaptive execution can produce useful real-rover exploration from onboard world feedback. Future work is safety-aware reward, separate sensor/safety monitors for low-cost hardware, and stronger pretrained visual encoders for room-level novelty.

References

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A. Appendix Placeholder

This appendix will contain extended run sheets, additional frame grids, reward-audit details, and repeated-run ablations. The main paper target is four pages before references and appendix.